

MATHEMATICAL INDUCTION

CSE 2213 – Discrete Mathematics

Course teacher: Muntaka Ibnath

LET'S CONSIDER THIS PROBLEM

Can you climb up staircases
of any height?

It will be tough, but **yes**, you can.

How can you prove it?

Seriously, we need to prove it???

YES, YOU HAVE TO PROVE IT.



ধ্যেৎ, ক্যান যে ১০০ তলা
সিঁড়ি দিয়ে উঠতে
গেসিলাম!!

CAN YOU CLIMB UP STAIRCASES OF ANY HEIGHT?

Only these two steps are to be proven:

? You can climb up the first step

? You can climb up the **next** step if you have already climbed up to **some** step

How???

You can climb up the **next** step if you have
already climbed up to **some** step



You can climb up the first step

MATHEMATICAL INDUCTION

A predicate is true for all values of its domain if –

? It is true for the beginning case

? If it is true for some case, then it is true for the next case

MATHEMATICAL INDUCTION

- A predicate $P(n)$ is true for all values $n \in \mathbb{Z}^+$ if –
 - $P(1)$ is true
 - Basis case
 - If $P(k)$ is true, then $P(k + 1)$ is true, where $k \in \mathbb{Z}^+$
 - Induction case

MATHEMATICAL INDUCTION

- A predicate $P(n)$ is true for all values $n \in N$ if –
 - Basis case: ???
 - Induction case: ???

MATHEMATICAL INDUCTION

- A predicate $P(n)$ is true for all prime numbers n if –
 - Basis case: ???
 - Induction case: ???

PROBLEM

Prove the summation formula of the following series for all positive integers n :

$$1 + 2 + 3 + 4 + \cdots + n = \frac{n(n + 1)}{2}$$

PROBLEM

Prove the summation formula of the following series for all nonnegative integers n :

$$1 + 2 + 2^2 + 2^3 + \dots + 2^n = 2^{n+1} - 1$$

PROBLEM

Prove the summation formula of the following series for all positive integers n :

$$1^2 - 2^2 + 3^2 - 4^2 + \dots + (-1)^{n-1}n^2 = (-1)^{n-1} \frac{n(n+1)}{2}$$

PROBLEM: PROVING DIVISIBILITY

- Prove that for all positive integer n , $n^3 - n$ is divisible by 3